ML-Based Student Academic Performance Prediction System

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Abstract*— *Student performance monitoring often depends on manual review of grades, attendance and student circumstances, which can delay support. This paper describes an ML-based Student Academic Performance Prediction System using the UCI Student Performance mathematics data and TabPFN. In the supplied project CSV, 395 records are labeled Good Performance when final grade G3 is at least 10 and At Risk otherwise; G3 is excluded from model inputs. Prior grades, attendance, study habits and selected support attributes are transformed into model features. The project materials describe stratified validation and a separate holdout evaluation. The project abstract reports 89.87% accuracy, 82.14% precision, 88.46% recall and 85.19% F1 on the holdout set. These are project-reported figures, not independently reproduced here. A Flask prototype provides individual prediction, a probability-derived risk score, a teacher dashboard and what-if scenarios. The paper documents the system scope, its evidence and limitations. The prediction is intended to inform educator review, not replace it.

Index Terms*— *educational data mining, student performance prediction, TabPFN, early warning, risk scoring, student support.

I. Introduction

Student outcomes are influenced by prior achievement, attendance, study habits and the learning environment. In a large cohort, manually reviewing these indicators can make it difficult to identify students who may need support before final results are available. Educational data-mining research has examined classification and regression models for student performance, but studies vary in population, target definition and evaluation protocol [1], [2].

A useful early-warning tool must state when a prediction can be made, exclude information that directly defines the outcome, and present uncertainty in a way that supports human review. A binary label alone may not tell an educator how much risk the model estimates or what aspects of a record warrant follow-up. The project addresses this practical workflow by combining a tabular classifier with a risk score, a teacher dashboard and selected what-if comparisons.

This paper makes four contributions:

- A leakage-aware binary target derived from final grade G3, with G3 excluded from model inputs.
- A TabPFN pipeline for a small UCI mathematics cohort with academic, attendance and support-related attributes.
- A Flask prototype that presents class predictions, risk scores, dashboard views and what-if scenarios.
- A transparent account of project-reported holdout metrics, implementation scope and unresolved evaluation limits.

Section II reviews related work. Section III describes the data and proposed workflow. Section IV reports implementation and available evaluation evidence. Section V discusses limitations; Section VI concludes and outlines next steps.

II. Related Work

A. Student Performance Prediction

Prior work applies decision trees, random forests, support-vector machines, logistic regression and neural models to student records. Reviews report frequent use of prior grades, demographic attributes and behavioral or learning-management data, while also highlighting variation in datasets and definitions of success [1], [2]. Cortez and Silva introduced the UCI secondary-school performance data for modeling final grades from student and school attributes [1].

B. Small-Data and Interpretable Models

The supplied literature includes spiking-neural prediction, belief-rule systems and small-data classifier comparisons [4], [5], [9]. Recent work expands the focus from predictive metrics to feature attribution, fairness auditing, calibrated warnings and counterfactual scenarios [6], [7]. These contributions are complementary: a strong classifier does not automatically provide a usable educator workflow, and an interpretable model does not establish that suggested actions cause improved outcomes.

C. Research Gap

The project gap is an integration and evaluation gap, not a claim that earlier studies lack risk prediction or explanation. The selected studies address different populations and tasks, so their scores cannot be ranked directly against this UCI cohort. The project instead connects a checkpoint-based risk estimate with educator review and input-level what-if scenarios.

TABLE I. RESEARCH GAPS AND PROJECT RESPONSE

Research gap	Project response
Prediction often separated from action	Risk score, dashboard and what-if review in one prototype.
Different datasets and target definitions	Define a G3 threshold for the 395-record math file; avoid cross-study score ranking.
Prediction time may be unclear	Use G1/G2 inputs; position output after those grade checkpoints.
Small-cohort validity is limited	Report scope and require calibration, subgroup and external validation.

III. Proposed Framework

A. System Overview

Fig. 1 summarizes the project workflow. A record available at the G1/G2 checkpoint is transformed into model inputs, passed to TabPFN and returned as a performance class and probability-based risk score. The dashboard and scenario tools support educator review; model output is advisory.

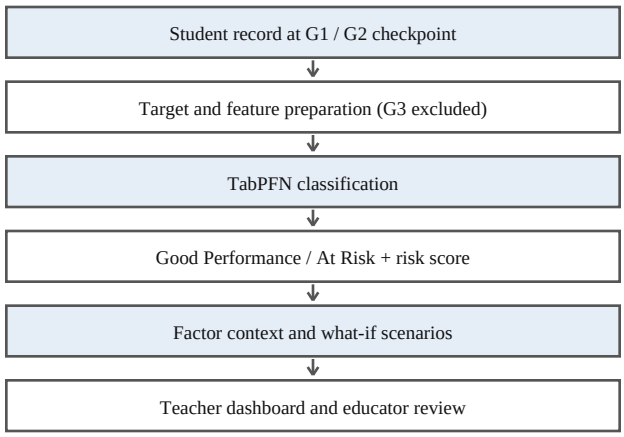

Fig. 1. Proposed prediction and educator-review workflow.

B. Target and Dataset

The working project file contains 395 mathematics records from the UCI Student Performance dataset [1]. Final grade G3 defines a binary target: Good Performance for G3 >= 10 and At Risk for G3 < 10. In the supplied file, the resulting classes contain 265 and 130 records, respectively. G3 is dropped from the predictors to prevent direct target leakage.

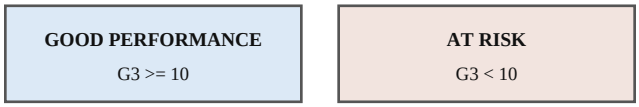

Fig. 2. Binary target definition from the final grade.

TABLE II. PROJECT DATA SUMMARY

Item	Description
Source	UCI Student Performance, mathematics subset [1]
Records	395 student records in the supplied working file
Target	G3 >= 10: Good; G3 < 10: At Risk
Predictors	11 selected source attributes plus 8 engineered features
Prediction point	After G1 and G2 are available

C. Feature Processing

Inputs include G1, G2, study time, absences, prior failures and selected demographic/support attributes. The project transformation creates 19 model columns, including grade averages/trends and support-related indicators. Since G1 and G2 are used, the model is not a start-of-term forecast. The risk score is 100 times the estimated probability of At Risk; the application groups records into low, moderate and high bands for review.

For the what-if interface, the archive includes four changes: G2 +2, absences -5, study time +1, and G2 +2 together with absences -5. These are model-input scenarios; they do not estimate causal intervention effects.

D. TabPFN and Training Protocol

TabPFN is a transformer-based method designed for small tabular classification problems [8]. The project uses it as the classifier for the binary target. The supplied abstract describes stratified five-fold cross-validation and a separate holdout set. The top-level training script uses an 80/20 stratified split (random state 42) and configuration selection on the training portion.

The archive contains two training entry points: a component trainer fixed at eight estimators and a top-level script that searches four, eight and sixteen. This difference should be reconciled before the reported result is treated as a canonical, reproducible model run.

E. Application and Technology

The project archive contains Python data transformation, TabPFN prediction/training and a Flask interface. The application supports an individual prediction view, probability-derived risk grouping, a teacher-facing dashboard and four what-if scenarios. The abstract also describes factor explanations and intervention recommendations; the supplied implementation materials do not establish their end-to-end behavior, so they are treated as capabilities requiring verification.

TABLE III. IMPLEMENTATION STACK

Layer	Technology / role
Data processing	Python, pandas, scikit-learn
Classifier	TabPFN for tabular classification
Web application	Flask; individual prediction and teacher dashboard
Evaluation	Stratified CV and holdout metrics, as described in project files

F. Project Evidence Status

The materials evidence four application functions in code and describe two additional functions in the abstract that still need end-to-end verification. The counts in Fig. 3 summarize document/code coverage only; they are not measures of model quality.

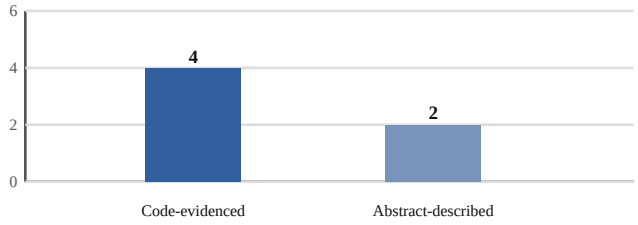

Fig. 3. Functions evidenced in project files versus abstract-described items to verify.

IV. Evaluation Plan and Results

A. Reported Holdout Results

The project abstract reports one holdout evaluation. Table IV and Fig. 4 reproduce those values as stated; the current paper does not independently rerun the model. Metric averaging details and the split manifest should be confirmed against the final training run before publication.

TABLE IV. PROJECT-REPORTED HOLDOUT METRICS

Metric	Reported value
Accuracy	89.87%
Precision	82.14%
Recall	88.46%
F1-score	85.19%

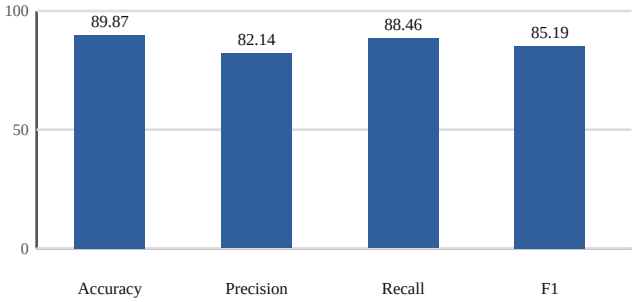

Fig. 4. Holdout metrics reported in the supplied project abstract.

The reported figures describe one project evaluation and should not be compared numerically with papers using different data, label thresholds or validation schemes. No confusion matrix, confidence interval or independent reproduction was supplied with the abstract.

B. Qualitative Scope Comparison

Fig. 5 summarizes explicit functional scope in selected approaches. It is a qualitative reading of the cited papers and project files, not an exhaustive review or effectiveness benchmark. The proposed project adds a student-level risk score, dashboard and what-if review to the prediction workflow; the distinction between code-evidenced and abstract-described functions is retained.

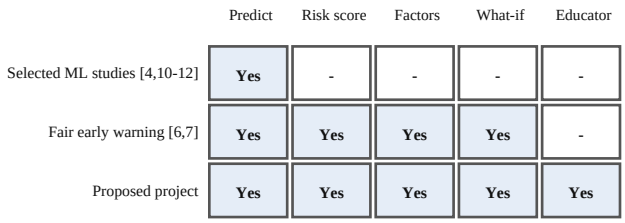

Fig. 5. Selected functional scope; qualitative, not a performance comparison.

C. Intended Use

At a grade checkpoint, an educator may review an At Risk probability, inspect the student record and compare a bounded what-if input change. The output can help prioritize a conversation or tutoring review. A simulated decrease in absences or increase in study time must not be presented as proof that the change will cause a higher grade.

V. Limitations and Threats to Validity

- The working cohort has 395 mathematics records; estimates may be unstable and should not be assumed to transfer to other schools, subjects or populations.
- The G3 threshold collapses a continuous grade into two classes and can hide variation near the pass/fail boundary.
- G1 and G2 make the model usable only after those checkpoints; it does not provide a beginning-of-term warning.
- The reported holdout metrics are reproduced from the project abstract and have not been independently rerun. Training entry points differ in estimator search settings.
- The supplied materials do not provide full calibration, subgroup fairness, confidence interval or external validation results.
- What-if outputs are conditional model predictions, not causal estimates of interventions. Explanations and recommendations require implementation and educator review.

Further work should reconcile the training scripts, record package/model versions and random seeds, publish the split and confusion matrix, and compare TabPFN with common baselines under the same protocol. Calibration, subgroup checks, a separate cohort and educator feedback are needed before operational use.

VI. Conclusion and Future Work

This paper described a TabPFN-based student-performance prediction prototype built around the UCI mathematics records. It defines Good Performance and At Risk from G3, excludes G3 from the predictors, and uses prior grades and selected student attributes to estimate risk. The supplied Flask application connects prediction with risk grouping, teacher dashboard review and what-if scenarios.

The contribution is the integration of a small-data classifier with a human-centered review workflow, not a claim of universal model superiority or proven intervention effect. The project abstract reports holdout metrics, but reproducibility, training-path consistency, calibration, fairness and external validity remain open. Future work should rerun a single documented pipeline, report class-wise results and probability calibration, validate on new cohorts, and assess whether explanations and recommendations are useful to educators.

Acknowledgment

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